

15/20

Nice!

GRA 1

4

[A] (2.)

$$m = \frac{\Delta R}{\Delta T}$$

$$= \frac{R_2 - R_1}{T_2 - T_1}$$

$$= \frac{77,5 - 68}{68 - 5}$$

$$= 0,47 \Omega/^{\circ}\text{C}$$

$$= 0,47 \Omega/^{\circ}\text{C}$$

2

(3)

$$y = mx + c$$

$$R = 0,47 \Omega/^{\circ}\text{C} T + 65,5 \Omega$$

2

(4) (i) at 5°C resistance = 68Ω

2

(ii) at 25°C resistance = $77,5 \Omega$

4 [B] (2)

$$n = \frac{\log(a_2) - \log(a_1)}{\log(T_2) - \log(T_1)}$$

$$= \log\left(\frac{a_2}{a_1}\right)$$

$$= \log\left(\frac{T_2}{T_1}\right)$$

$$= \log\left(\frac{5}{3,5 \times 10^{-1}}\right)$$

$$= \log\left(\frac{10}{3,5 \times 10^{-1}}\right)$$

$$= 0,96 \text{ au./yo too high}$$

1

0

4)

$$a = k T^n$$

$$\log(a) = \log(k) + 0,96 \log(T)$$

$$\log(a) = \log(2 \times 10^{-1}) + 0,96 \log(T)$$

0

5)

Yes. It's linear graph

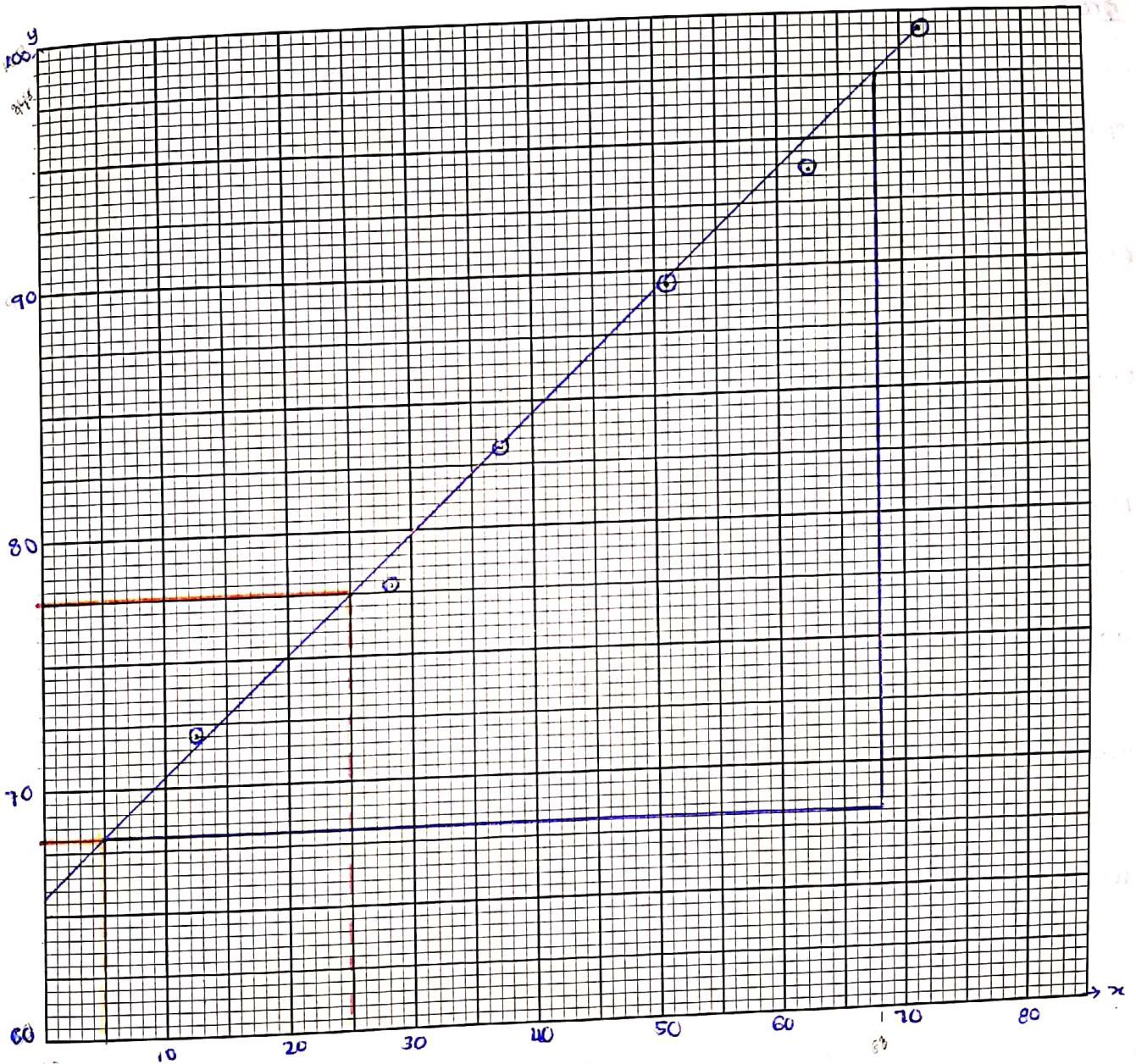
Explain

DATE: 12/02/19

EKSP/ EXP: 1

The resistance (R) of a homogeneous wire at different temperatures ($T \rightarrow x$)

Resistance, R (Ω)



Temperature, T ($^{\circ}\text{C}$)

4

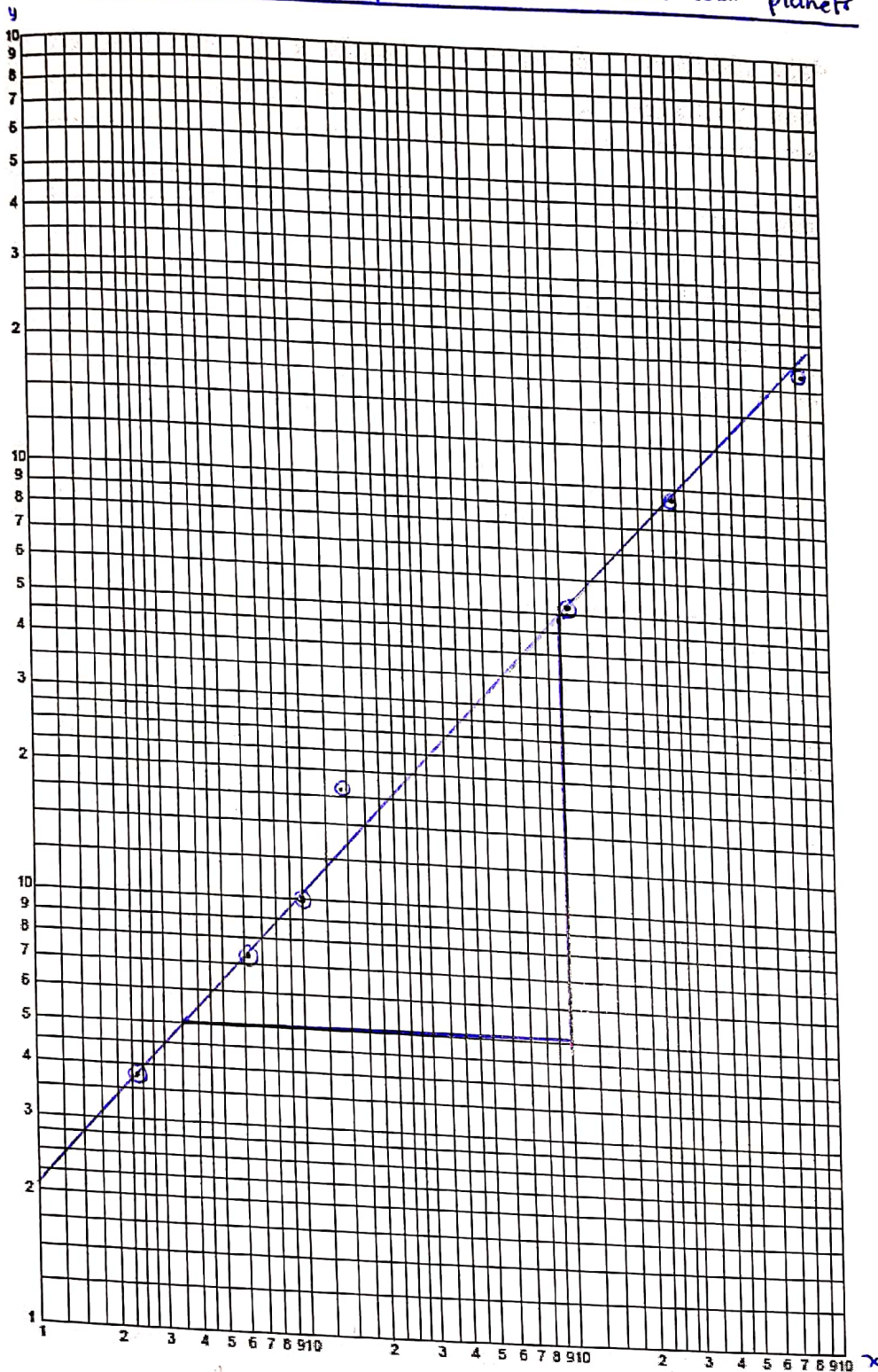
GRA 1

DATE: 12/02/19

EKSP/EXP: 1

Semi-major axes and periods of orbit of some solar planets

semi-major axes of orbit, a (a.u.)



Time, T (yr)

GRA 17

4

2
4GRA 2

(natural logarithm of count rate)

2) Decreasing activity of radioactive sample

Time, t (minutes)	The natural logarithm of the count rate ($\ln(R)$)
0,0	6,68
5,0	5,99
10,0	5,32
15,0	4,65
20,0	3,91

4) $y_0 = 6,68$

3

$$m = \alpha = \frac{18,75 - 1,25}{4,1 - 6,5} \quad m = \alpha = \frac{4,1 - 6,5}{18,75 - 1,25}$$

$$= -0,14 \text{ min}^{-1}$$

5)

$$\ln(R) = y_0 + \alpha t$$

$$= 6,68 - 0,14 \text{ min}^{-1} t$$

$$\approx 6,68$$

6)

$$R_0 = e^{6,68}$$

$$= 796,32 \text{ counts/s}$$

$$\alpha = -\lambda$$

$$\lambda = -\alpha$$

$$= -(-0,14 \text{ min}^{-1})$$

$$= 0,14 \text{ min}^{-1}$$

2

$$y\text{-intercept} = 6,68$$

(20/20)

Like a Boss!!

2

$$(7) \quad R = R_0 e^{-\lambda t} \\ = 796,32 \text{ counts/s } e^{-0,14 \text{ min}^{-1} t}$$

3

$$(8) \quad R = R_0 e^{-\lambda t} \\ \frac{R_0}{2} = R_0 e^{-\lambda T_{1/2}} \\ \frac{1}{2} = e^{-\lambda T_{1/2}} \\ \ln\left(\frac{1}{2}\right) = \ln(e^{-\lambda T_{1/2}})$$

$$\ln(1) - \ln(2) = -\lambda T_{1/2}$$

$$0 - \ln(2) = -\lambda T_{1/2}$$

$$-\ln(2) = -\lambda T_{1/2}$$

$$\ln(2) = \lambda T_{1/2} \rightarrow$$

2

$$(9) \quad \ln(2) = \lambda T_{1/2} \rightarrow T_{1/2} = \frac{\ln(2)}{\lambda}$$

$$= \frac{0,14}{0,69}$$

$$= \frac{0,693}{0,14}$$

$$= 4,95$$

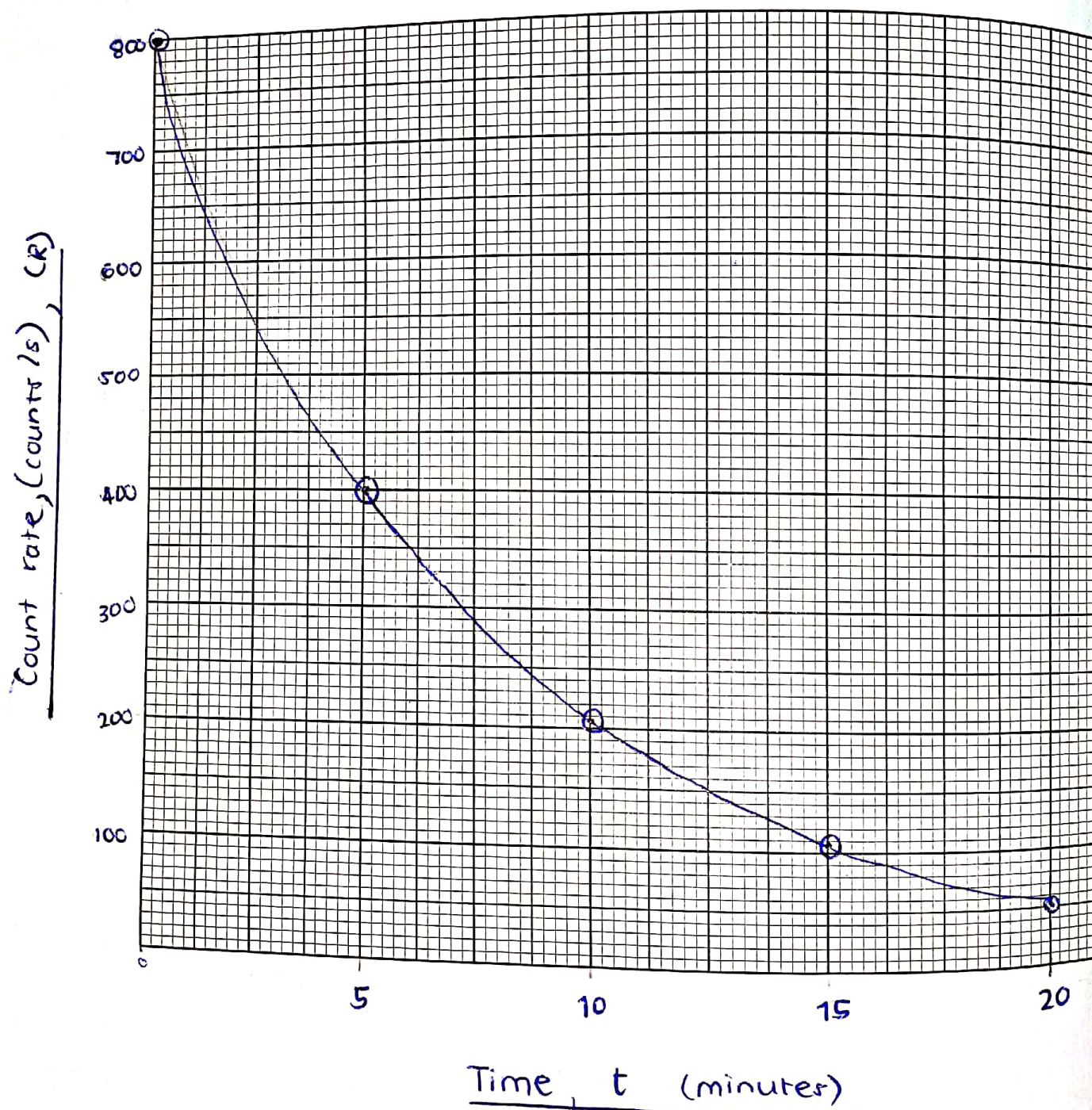
$$= 5,0 \text{ minutes} \rightarrow$$

DATE: 19/02/2019

EKSP/EXP: 2

(count rate)

The decreasing activity of a radioactive sample over a period of 20 minutes



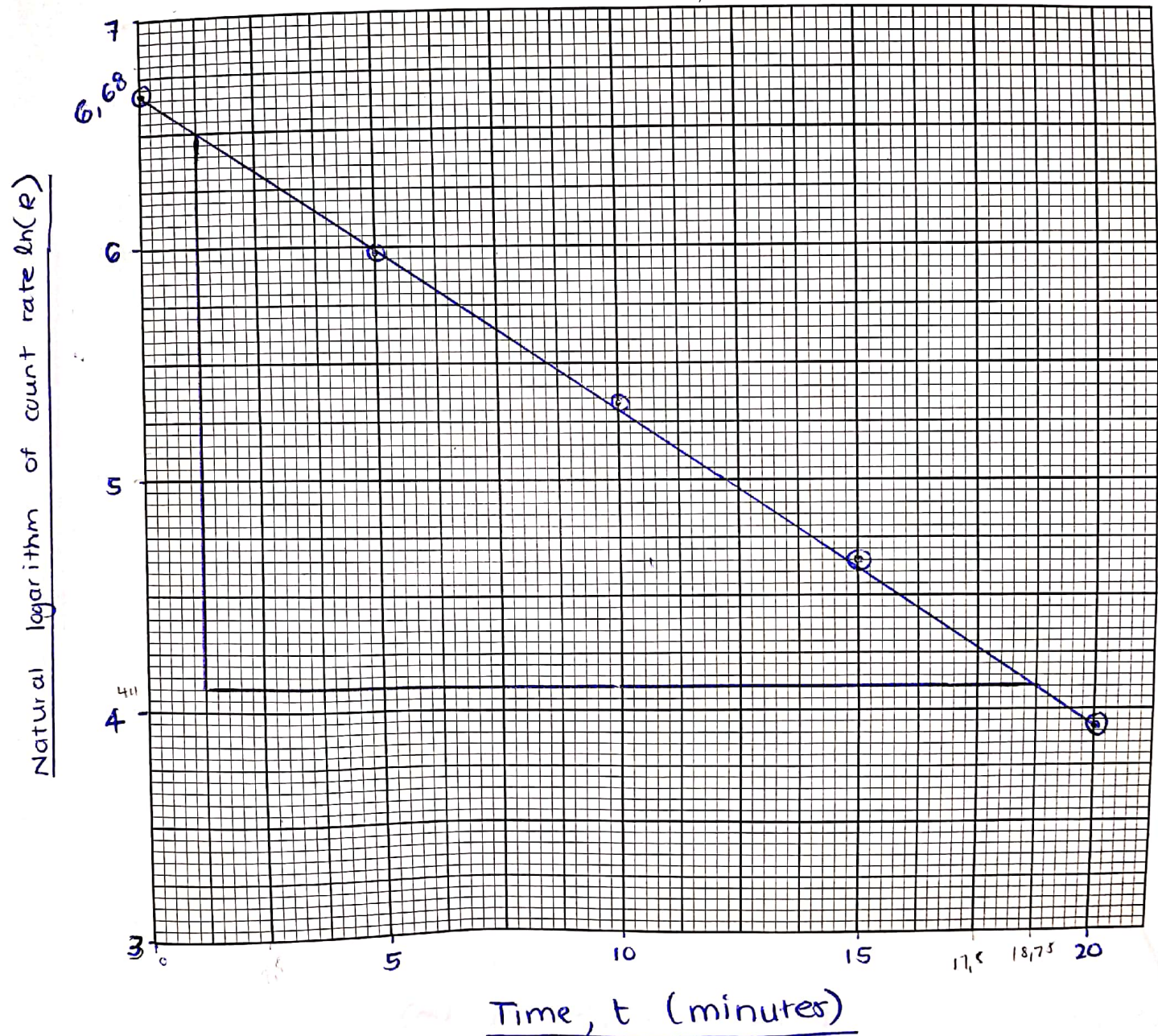
(2)

GRA 2

DATE: 19/02/2019

EKSP/EXP: 2

The decreasing activity of the natural logarithm of count rate of a radioactive sample over a period of 20 minutes



4

GRA 3

20/20

V. Good

Heading

MET - measurement and units

Objectives

Learn how to :

- * use the vernier calliper with the necessary dexterity and accuracy to make measurements
- * use an electronic scale to determine the mass of an object. (1)
- * convert measurements to SI units
- * round calculated values off to the correct number of significant figures.

[A] Significant Figures

- (a) $4,3 \times 10^3 \text{ km} \rightarrow 2$ significant figures ✓ (3)
- (b) $0,0160 \text{ s} \rightarrow 3$ significant figures ✓
- (c) $200 \text{ g} \rightarrow 1$ significant figure ✓
- (d) 2 cm has 1 significant figure, but $2,0 \text{ cm}$ has 2 significant figures making $2,0 \text{ cm}$ a more accurate measurement with (1)

e) $\rho = \frac{m}{V}$

$$m = \rho V$$

$$= (1,33 \times 10^3 \text{ kg/m}^3) \times (0,35 \text{ m} \times 0,057 \text{ m} \times 1,000 \text{ m})$$

$$= 26,5335 \text{ kg}$$

$$\approx \underline{26,53 \text{ kg}}$$

$$\approx \underline{27 \text{ kg}}$$
 ✓ (1)

$$\begin{aligned}
 (f) \quad V &= \frac{4}{3} \pi r^3 \\
 &= \frac{4}{3} \times \pi \times (6,5 \times 10^{-3} \text{ m})^3 \\
 &= \underline{1,15 \times 10^{-6} \text{ m}^3}
 \end{aligned}$$

$$\rho = \frac{m}{V}$$

$$m = \rho V$$

$$= (1,18 \times 10^3 \text{ kg/m}^3)(1,150 \times 10^{-6} \text{ m}^3)$$

$$= 1,357 \times 10^{-3} \text{ kg}$$

$$\approx 1,36 \times 10^{-3} \text{ kg}$$

$$\approx \underline{1,36 \text{ g}}$$

[B] Vernier caliper readings

1.) 24,5 mm ✓

2) 17,5 mm ✓

3) 0,76 mm ✓

4) 17,46 mm ✓

5) 9,90 mm ✓

6) 12,70 mm ✓

[C] Density determination

$$1) \quad V = hw^2 - h(w-2t)^2 \\ = h(w^2 - (w-2t)^2)$$

2) Dimensions of tube

	height, h (m)	width, w (m)	thickness, t (m)
	$8,94 \times 10^{-2}$	$3,80 \times 10^{-2}$	3×10^{-3}
	$8,95 \times 10^{-2}$	$3,80 \times 10^{-2}$	$2,76 \times 10^{-3}$
	$8,94 \times 10^{-2}$	$3,81 \times 10^{-2}$	$2,9 \times 10^{-3}$
AVERAGE	0,09	0,04	$2,89 \times 10^{-3}$

(4)

$$3) \quad m = 99,3g \\ = 0,0993 \text{ kg}$$

$$V = h(w^2 - (w-2t)^2) \\ = 0,09 \text{ m} (0,04 \text{ m}^2 - (0,04 \text{ m} - 2(2,89 \times 10^{-3} \text{ m}))^2) \\ = 3,86 \times 10^{-5} \text{ m}^3$$

$$\rho = \frac{m}{V} \\ = \frac{0,0993 \text{ kg}}{3,86 \times 10^{-5} \text{ m}^3} \\ = 2571,92 \text{ kg/m}^3 \\ \approx 2,57 \times 10^3 \text{ kg/m}^3$$

(2)

4) The material used was aluminium ~~(which has)~~

CONCLUSION

In this experiment we learnt ~~how~~ to accurately make measurements using the vernier calliper, how to measure mass using an electronic scale, how to convert to SI units and round off values to the correct number of decimal places.

$$3 + 11 = 14$$

PAR - Projectile motion

Objectives

- Calculate y -coordinates of a horizontally launched projectile as it crosses points along the x -axis and thereby make assumptions about the projectile
- Measure projectile's actual y -coordinates and compare calculated points to measured ones.
- * Determine how the path followed by a projectile comes about.

First set of measurement

TABLE 1 :

Height ball reaches

Height, h (m)	2.24	2.24	2.25
-----------------	------	------	------

$$h_{avg} = \frac{2.24 + 2.24 + 2.25}{3} \text{ m}$$

$$h_0 = \frac{6.73}{3} \text{ m}$$

$$= 2.24 \text{ m}$$

$$E_p = E_k$$

$$mgh_0 = \frac{1}{2}mv_0^2$$

$$v_0^2 = 2gh$$

$$v_0 = \sqrt{2gh}$$

$$= \sqrt{2(9.79 \text{ m/s}^2)(2.24 \text{ m})}$$

$$= 6.62 \text{ m/s}$$

Q.1

$$V_0 = \frac{x_1}{t_1}$$

$$t_1 = \frac{x_1}{V_0}$$

$$= \frac{35 \times 10^{-2} \text{ m}}{6.62 \text{ m/s}}$$

$$= 0.0529 \text{ s} \approx 0.053 \text{ s}$$

$$V_0 = \frac{x_2}{t_2}$$

$$t_2 = \frac{x_2}{V_0}$$

$$= \frac{70 \times 10^{-2} \text{ m}}{6.62 \text{ m/s}}$$

$$= 0.1057 \text{ s} \approx 0.106 \text{ s}$$

Q.2

$$V_{0y} = 0 \text{ m/s}$$

Yes, the downward speed of the ball will change as the ball motion progresses, as it would be dependant on only the acceleration and time

$$\text{ie. } V_y = v_{iy} + a_y t$$

$$= 0 + a_y t$$

$$= a_y t$$

$\therefore$ It would decrease as the motion progresses as $a = -9.78 \text{ m/s}^2$

Q.3

$$y_f = y_i + V_{0y} t - \frac{1}{2} g t^2$$

$$y_f - y_i = V_{0y} t - \frac{1}{2} g t^2$$

$$\text{Let } d = y_f - y_i$$

$$\therefore d = 0 - \frac{1}{2} g t^2$$

$$d = -\frac{1}{2} g t^2 \quad \text{--- (B)}$$

$$\text{where } g = -9.78 \text{ m/s}^2$$

2nd set of measurements

Table 2 :

Table? (-1)

$x \times 10^{-2}$ (m)	time, t (s)	distance, d (m)	y_c (m)	y_e (m)
35	0.0529	0.0137	0.8463	0.818
70	0.106	0.0549	0.8051	0.808
105	0.159	0.124	0.736	0.732
140	0.211	0.218	0.642	0.633
175	0.264	0.341	0.519	0.530
210	0.317	0.491	0.369	0.405
245	0.370	0.669	0.191	0.146
280	0.423	0.875	0	0

$$h_i = 0.86 \text{ m}$$

Conclusion: The path followed by a projectile is a parabolic path at the vertical position is only influenced by constant acceleration (g) while the horizontal velocity is usually constant. There is motion in both the ie. ~~x~~ x = horizontal and vertical directions.

There is an inverse relationship between the motion in the 2 directions

Q.4

- converting and systematic errors
- error in recording the measurements of setting up the measuring instrument

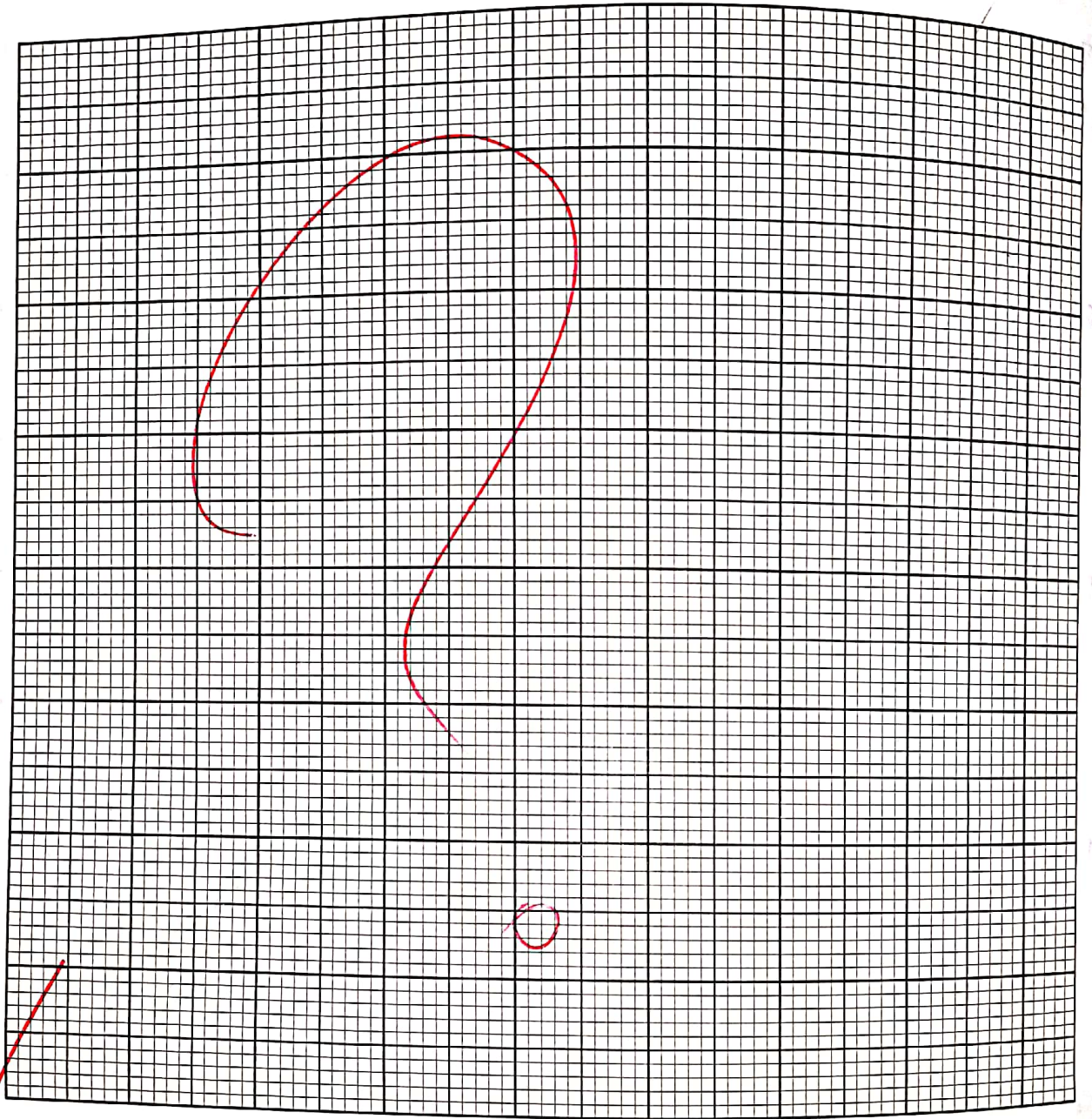
2

DATE: 05/03/2019

EKSP/EXP: PAR 4

Relationship of motion of the object in the vertical and horizontal direction

Vertical displacement (m)



Horizontal displacement (m)

ELA - Elasticity

Objectives

- To investigate the elasticity of a steel spring and rubber band and to graphically illustrate the relationship between the ^{load} force and extension of a steel spring
- To determine the force constant k using Hooke's Law

Results

Table of extension against load

Mass (kg) $\times 10^{-3}$	y' (m) $\times 10^{-3}$	y (m) $\times 10^{-3}$	Force/Load (N)
50	4	4	0.489
100	15	15	0.978
150	31	31	1.467
200	45	45	1.956
250	59	59	2.445
200	45	45	1.956
150	31	31	1.467
100	18	18	0.978
50	6	6	0.489

8. The spring obeys Hooke's Law as the graph is a straight line which indicates that $w \propto y$

9. ~~8/1~~ $m = \frac{\Delta y}{\Delta x}$
 $= \frac{(59 \times 10^{-3} - 4 \times 10^{-3}) \text{ m}}{(2.43 - 0.5) \text{ N}}$
 $= 0.027 \text{ m/N}$

Handwritten signature

$$y = mx + c$$

$$4 \times 10^{-3} = (0.027 \frac{m}{N})(0.489) + c$$

Point $(4 \times 10^{-3}, 0.489)$

$$c = -9.2 \times 10^{-3}$$

$$\therefore y = 0.027 \frac{m}{N} x_{(N)} - 9.2 \times 10^{-3} (m)$$

10) $y = \frac{x}{k}$

k is the inverse of the gradient of the graph

11) $k = \frac{1}{0.027 m/N}$
 $= 37.037 N/m$

[B]

Table of extension against load

Mass (kg) $\times 10^{-3}$	y_1 (m) $\times 10^{-3}$	y (m) $\times 10^{-3}$	Force/Load (N)
50	9	9	0.489
100	15	15	0.978
150	28	28	1.467
200	44	44	1.956
250	64	64	2.445
200	52	52	1.956
150	37	37	1.467
100	24	24	0.978
50	13	13	0.489

The elastic band did not return to its original length.
It stretched out.

Questions

1) Yes there are differences. The rubber bands elasticity is smaller than the spring and it doesn't obey Hooke's Law whilst the spring does obey it.
The spring returns to its original length and is more rigid.
The value of k of the spring remains constant while the value of k of the rubber band changes.

2) Hooke's Law can be used to describe the spring, not the rubber band. The band doesn't obey it as load is not directly proportional to extension.

3) The spring. It is more rigid and the rubber band deforms. The spring obeys Hooke's Law.

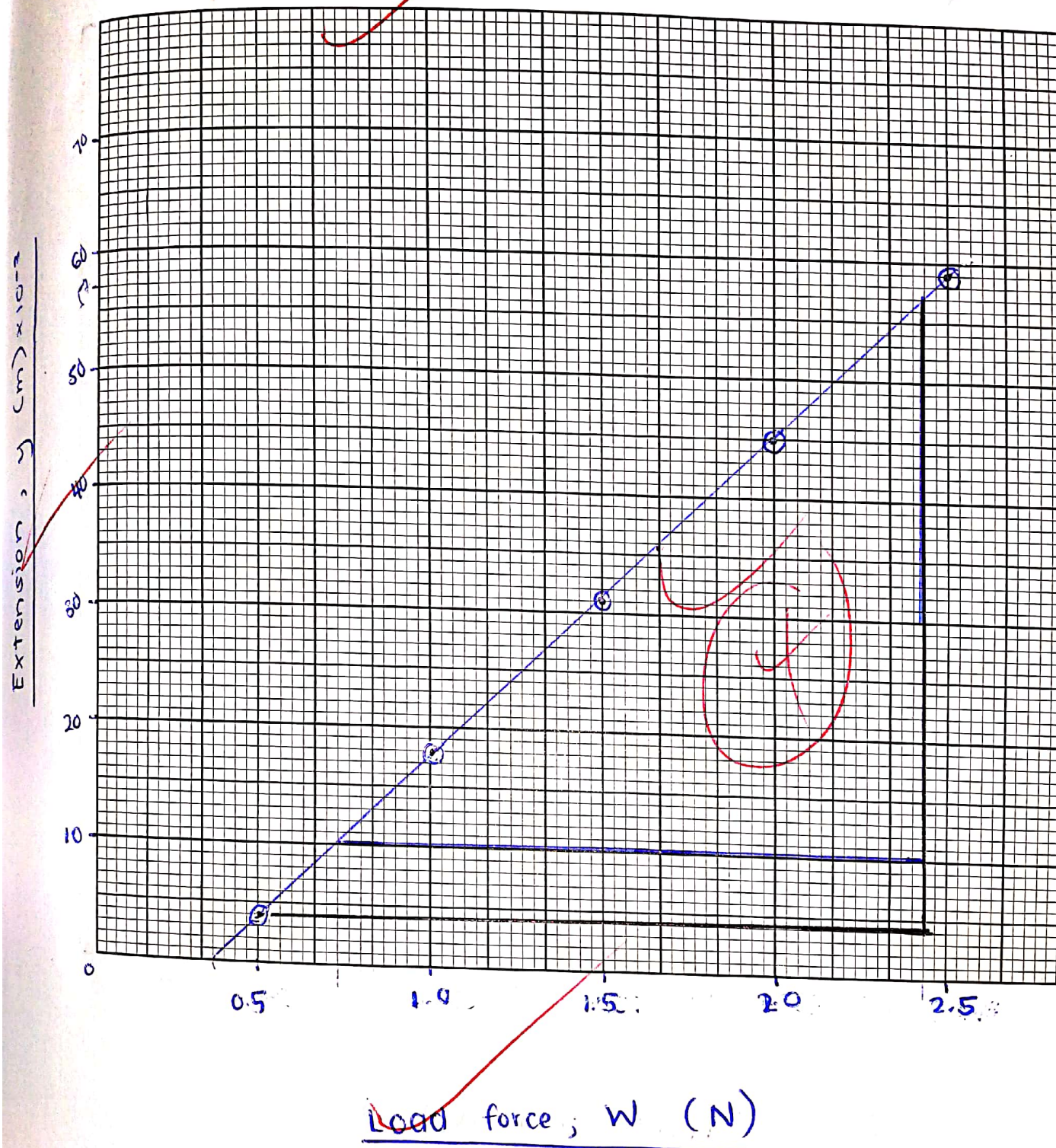
Conclusion

Rubber band is less elastic than the spring. The spring obeys Hooke's Law and the rubber band doesn't.
The force constant k can be calculated using the inverse of the gradient¹³ of the graph.

DATE: 02/04/2019

EKSP/EXP: 5

Graph of extension of spring (m) as a function of the load force (N)

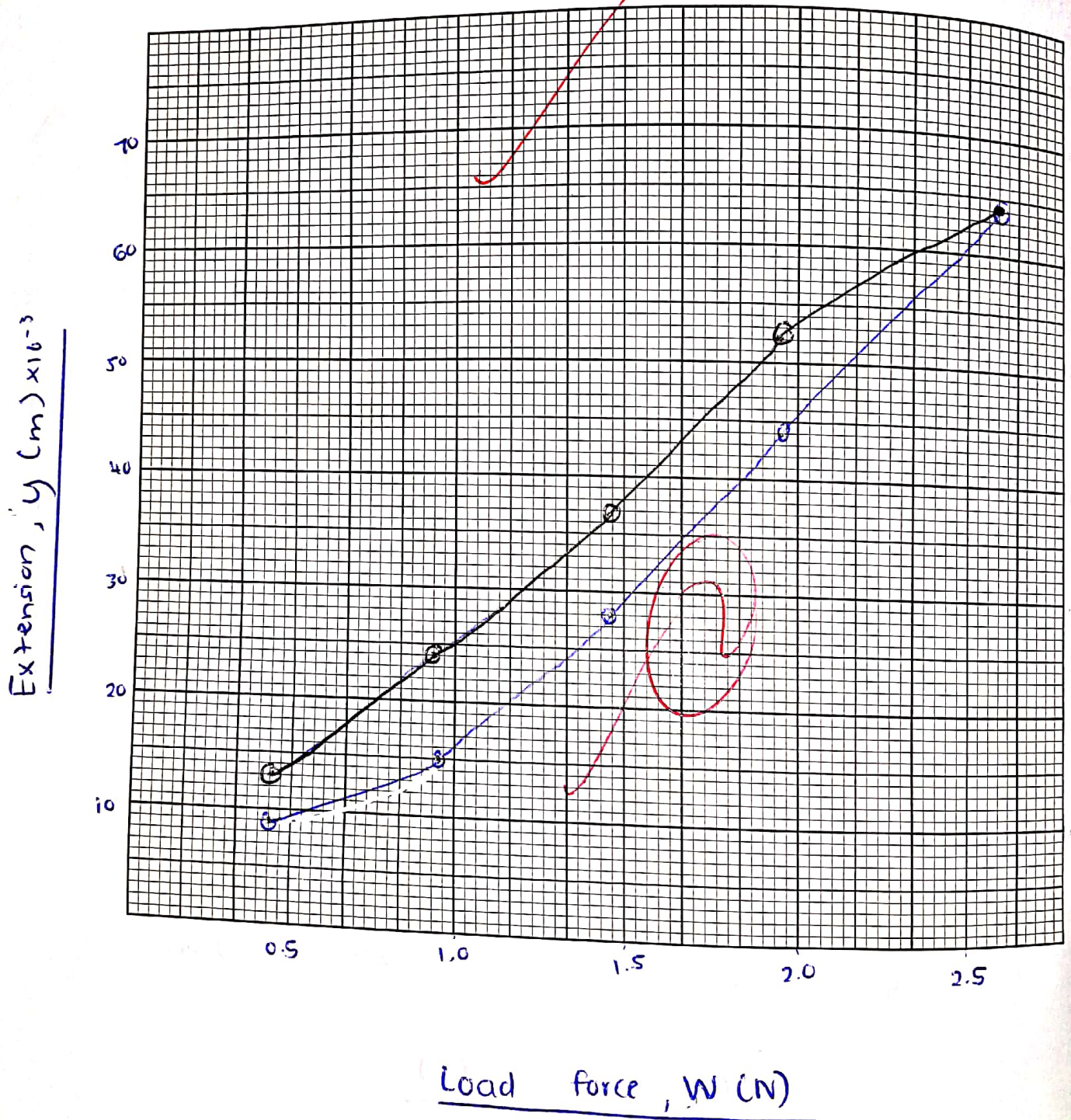


(2.43, 57)

DATE: 02/04/19

EKSP/EXP: 5

Graph of extension of rubber band spring (m) as a function of the load force (N)



— load
— unload

13/2

SHM - Simple Harmonic Motion

① AIM : To investigate the oscillations of a mass piece hanging on a spring and to determine if it performs simple harmonic movement

② Determine k

	y (mm)	F _g (N)
1.	20	+ 0.15
2.	40	+ 0.38
3.	60	+ 0.51

$$F_g = ky$$

$$k = \frac{F_g}{y}$$

$$k_1 = \frac{F_g}{y} = \frac{0.15 \text{ N}}{20 \text{ mm}} = 7.5 \times 10^{-3} \text{ N/mm}$$

$$k_2 = \frac{F_g}{y} = \frac{0.38 \text{ N}}{40 \text{ mm}} = 9.5 \times 10^{-3} \text{ N/mm}$$

$$k_3 = \frac{F_g}{y} = \frac{0.51 \text{ N}}{60 \text{ mm}} = 8.5 \times 10^{-3} \text{ N/m}$$

$$k_{\text{avg}} = \frac{(7.5 \times 10^{-3} + 9.5 \times 10^{-3} + 8.5 \times 10^{-3}) \text{ N/mm}}{3}$$

$$= 8.5 \times 10^{-3} \text{ N/mm} = 0.0085 \text{ N/mm}$$

③ Results of the oscillation (Table of Time vs force and calculated y_e)

time, t (s)	Force, F _g (N)	y _e (mm)	y _t (mm)	Δy (mm)
0.134	1.924	226.35		
0.268	1.580	186.12		
0.402	-1.175	138.24		
0.536	-1.1016	119.53		
0.670	-1.173	138		
0.804	-1.553	102.71		
0.938	-1.912	224.94		
1.072	-1.993	234.47		

4) Determining ω and y_m

4.1) Determine Period (T) :

$$T_1 = 1.543_s - 0.536_s = 1.007_s$$

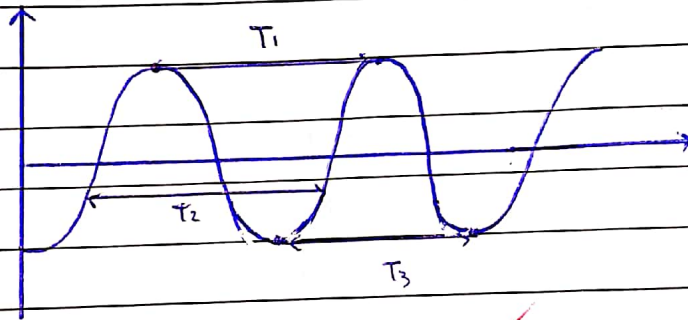
$$T_2 = 2.209_s - 1.029_s = 1.18_s$$

$$T_3 = 1.339_s - 0.373_s = 0.966_s$$

$$T_{avg} = \frac{1.007 + 1.18 + 0.966}{3}$$

$$= 1.051_s$$

4.2) Rough Sketch



4.3) $\omega = \frac{2\pi}{T}$

$$= \frac{2\pi \text{ rad}}{1.051_s}$$

$$= 5.978 \text{ rad/s}$$

$$\omega^2 = \frac{k}{m}$$

$$= \frac{8.5 \text{ N/m}}{258.16 \times 10^{-3} \text{ kg}}$$

$$= 32.93$$

$$\omega = \sqrt{32.93}$$

$$= 5.74 \text{ N/kg} \cdot \text{m}$$

$$= 5.74 \text{ rad/s}$$

$$\omega_{avg} = \frac{5.978 + 5.74}{2}$$

$$= 5.86 \text{ rad/s}$$

$$44) \quad F_{\max} = -1.016 \text{ N}$$

$$F_{\min} = -2.004 \text{ N}$$

$$y_m = \frac{F_{\max} - F_{\min}}{2k}$$

$$= \frac{[-1.016 - (-2.004)] \text{ N}}{2[(8.5 \times 10^{-3}) \text{ N/mm}]}$$

$$= 58.12 \text{ mm}$$

$$5) \quad y(t) = y_m \cos(\omega t + \phi)$$

$$y(0) = y_m \cos(\phi)$$

$$\cos \phi = 4.15$$

$$\phi = 1.99 \text{ rad}$$

$$y_0 = \frac{F_0}{k}$$

$$= \frac{2.052}{8.53 \times 10^{-3}}$$

$$= 241.41 \text{ mm}$$

Conclusion

Δy is small which indicates that the mass piece performed simple harmonic motion

CLM - Linear Momentum

Aim / Objective

To determine whether the linear momentum of the two discs is conserved before and after the collision.

Results

Table of results (measurements before and after collision)

	Before collision		After collision	
	STRIKER	TARGET	STRIKER	TARGET
Mass (kg)	14.6×10^{-3}	5.70×10^{-3}	14.6×10^{-3}	5.70×10^{-3}
Δs (m)	10.4×10^{-2}	0	4.52×10^{-2}	11.6×10^{-2}
Velocity (m/s)	2.60	0	1.13	2.90
Momentum (kg·m/s)	0.0380	0	0.0165	0.0165
Angle ($^{\circ}$ - degrees)	4.1	0	-14.4	21.1

$e = 0.045$

Discussion of results

Q.1) No, the discs don't travel with a constant speed along their tracks.

Q.2) No. The velocity was not conserved and the system was not an isolated system. There were external forces such as friction present.

Q.3) Friction

Q.4) The system was not an isolated system and therefore

the velocities change due to external forces such as friction. Therefore using velocities just before and after gives most accurate results

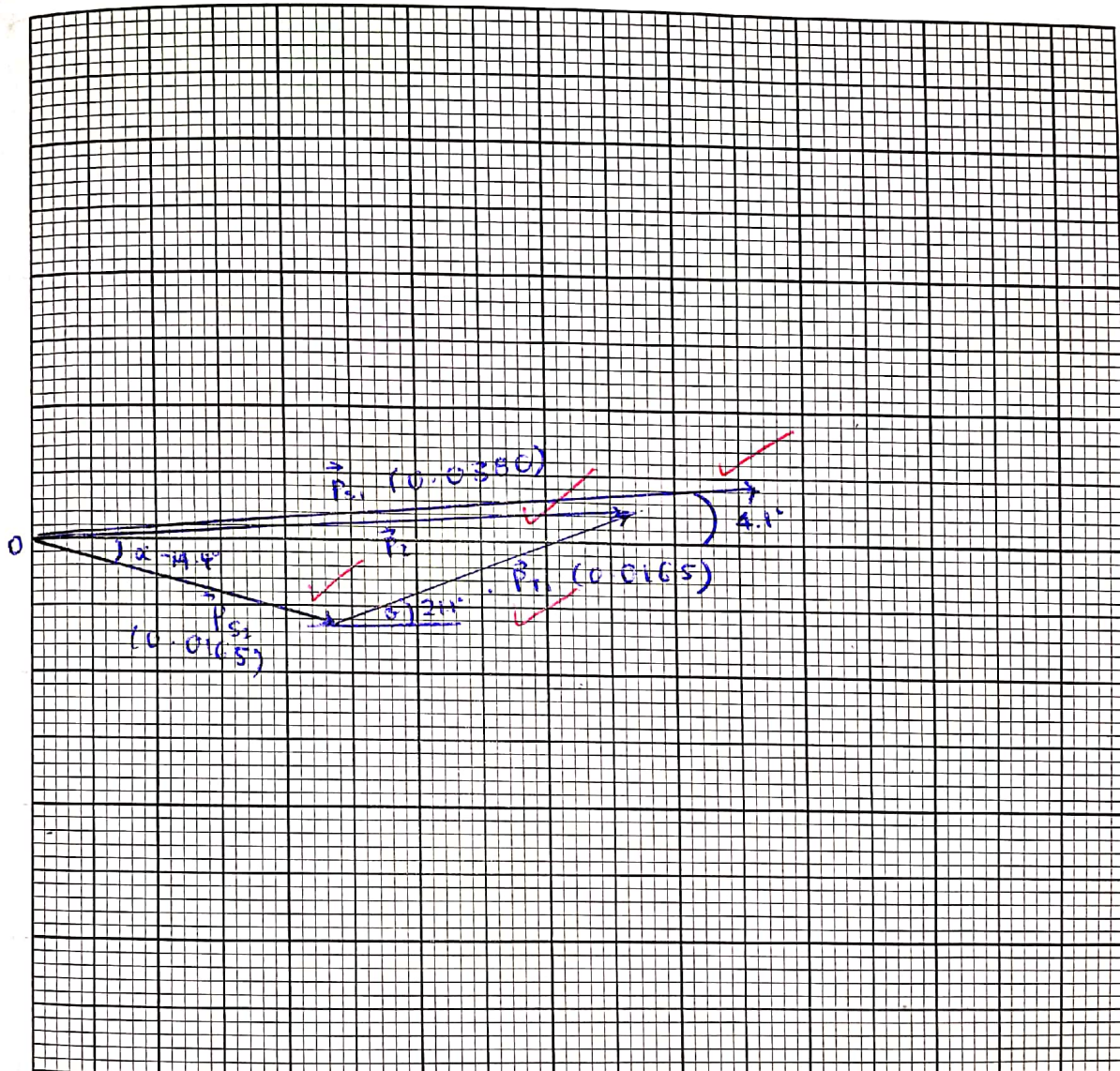
Conclusion 5) No. Momentum was not conserved as P_1 and P_2 were not the same lengths/magnitudes. The system was not an isolated system. ✓

5

DATE: 16/04/2019

EKSP/EXP: 7
CLM

Vector diagram of the momentum of the striker and target



Scale factor : $\frac{10\text{cm}}{0.0380\text{ kg}\cdot\text{m/s}}$

$= \frac{263.16\text{ cm}}{\text{kg}\cdot\text{m/s}}$ ✓

$1\text{cm} = 100\text{ kg}\cdot\text{m/s}$
 $1\text{m} =$

GRA 7

HEA - Heat

Aim:

To investigate the relationship between the amount of ~~water~~ heat added to a certain amount of water and the change in its temperature.

Results

$$m_w = (109.8 - 2.9) \times 10^{-3} \text{ kg} = 106.9 \times 10^{-3} \text{ kg} = 0.107 \text{ kg}$$
$$I_i = 2.60 \text{ A} \quad I_f = \quad I_{\text{avg}} = \frac{2.6 + 2.5}{2} = 2.55 \text{ A}$$
$$V_i = 3 \text{ V} \quad V_f = 3 \text{ V} \quad V_{\text{avg}} = \frac{3 + 3}{2} = 3 \text{ V}$$

Table representing relationship between temperature ($^{\circ}\text{C}$) and heat gained (J) over period of time (ie 5 minutes)

Time, t (min)	Temperature, T , ($^{\circ}\text{C}$)	Change in temp, ΔT , ($^{\circ}\text{C}$)	Heat gained, Q_w (J)
0	31.6 25.6	0	0
1	33.6 25.7	2.00 2.00	459
2	34.4 25.8	2.80 2.80	918
3	35.1	3.50	1377
4	35.7	4.10	1836
5	36.4	4.80	2295

$Q = VIt$

4)

$$y = mx + c$$
$$\Delta T = m Q_w + c$$

5)

$$m = \frac{\Delta y}{\Delta x}$$
$$= \frac{(4.5 - 1.6)^{\circ}\text{C}}{(2400 - 180) \text{ J}}$$

$$= 1.53 \times 10^{-3} \text{ }^{\circ}\text{C/J}$$

$$c = 1.18$$

$$6) \Delta T (^{\circ}\text{C}) = (1.53 \times 10^{-3} ^{\circ}\text{C/J}) Q_w + 1.18 ^{\circ}\text{C} \quad \checkmark$$

$$c) \Delta T = \frac{1}{m_w c_w} \cdot Q_w$$

$$7) \frac{1}{m_w c_w} = 1.53 \times 10^{-3} ^{\circ}\text{C/J}$$

$$8) 1.53 \times 10^{-3} ^{\circ}\text{C/J} = \frac{1}{0.107 + c_w}$$

$$0.107 + c_w = 652.94$$

$$c_w = \cancel{652.83} \text{ [J/kg} \cdot ^{\circ}\text{C]} \quad \checkmark$$

$$(c_w = 652.83 \text{ [J/kg} \cdot ^{\circ}\text{C]}) \quad \checkmark$$

$$1.53 \times 10^{-3} ^{\circ}\text{C/J} = \frac{1}{0.107 + c_w}$$

$$0.107 + c_w = \frac{1}{1.53 \times 10^{-3}}$$

$$c_w = \cancel{652.83}$$

$$9) \% \text{ difference} = \frac{\text{actual} - \text{determined}}{\text{determined}} \times 100 \% \quad \checkmark$$

$$= \frac{(4200 - 652.83)}{652.83} \times 100 \%$$

$$= 5.43 \% \quad \times$$

10) $\rightarrow$ incorrect rounding off

$\rightarrow$ not an isolated system

$\rightarrow$ heat could have been lost - resulting in an incorrect/inaccurate value of ΔT

11) Conclusion

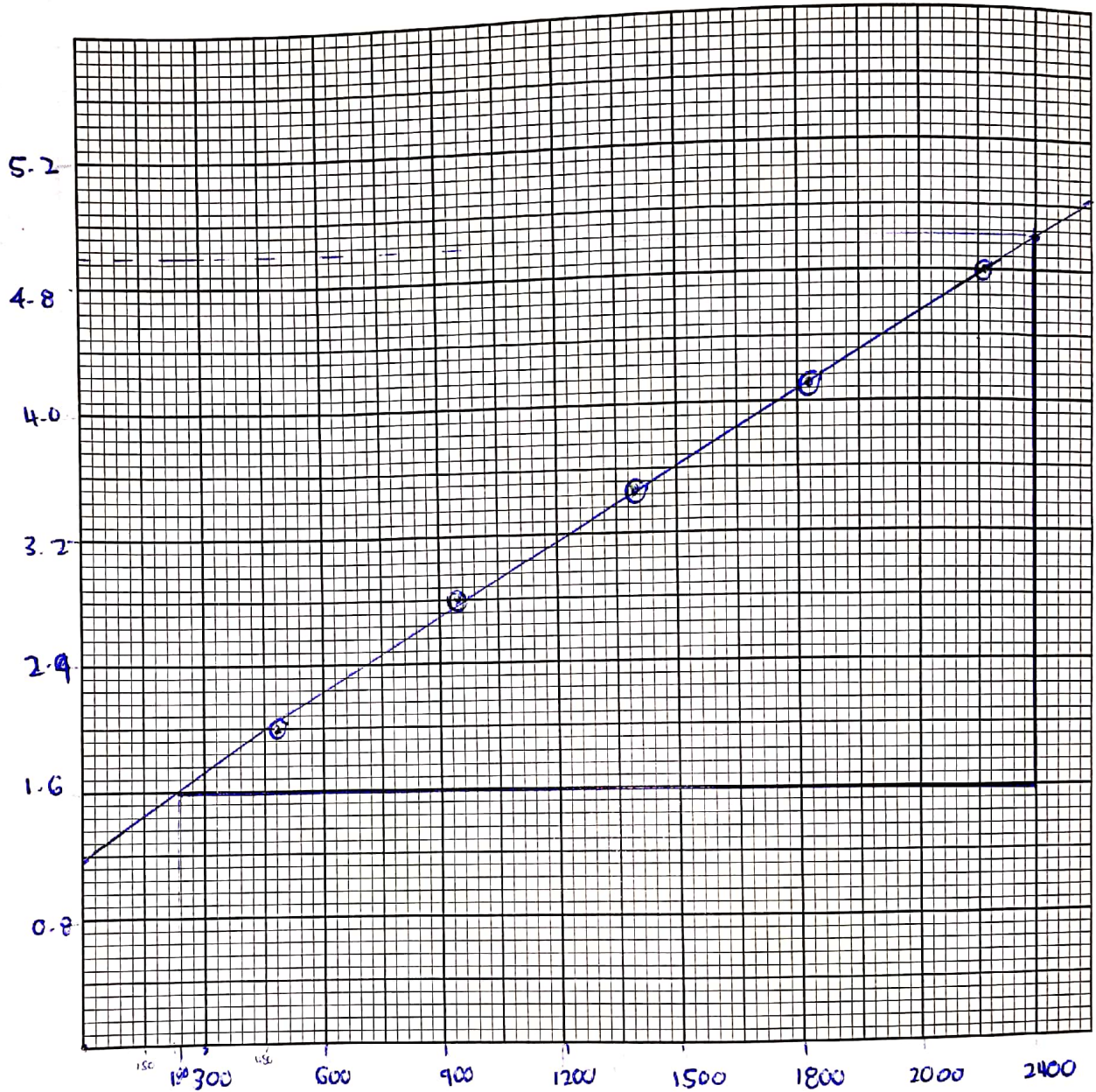
The amount of heat added to water is directly proportional to the change in its temperature.

DATE: 23/04/2019

EKSP/EXP: 8
HEA

The relationship between the amount of heat added to the water and the change in temperature of the water

Temperature change of water, ΔT , $^{\circ}\text{C}$



Heat added to the water, Q_w (J)

20/20

ROI - Rotational inertia

Aim

To experimentally determine the rotational inertia, I , or moment of inertia of a flat disc rotating about a perpendicular axis through its center

Equations

• Applied torque

$$\tau = rT$$

$$= rm(g - a)$$

$$\approx rmg \quad \dots \quad (1)$$

$$r = 50 \text{ mm}$$

$$r = \frac{50}{2} \text{ mm}$$

$$= 0.025 \text{ m}$$

• Angular acceleration

$$\Delta\theta = 2\pi N$$

$$= \frac{1}{2} \alpha t^2 \quad \dots \quad (2)$$

• Newton II

$$\sum \tau = I\alpha$$

$$\tau - \tau_0 = I\alpha$$

$$\tau = I\alpha + \tau_0$$

from graph

$$\tau_0 = 0.6 \times 10^{-3} \text{ N}\cdot\text{m}$$

$$m = I = \frac{\Delta\tau}{\Delta\alpha} = \frac{(12 \times 10^{-3} \text{ N}\cdot\text{m}) - (2 \times 10^{-3} \text{ N}\cdot\text{m})}{1.75 \text{ (rad/s}^2) - 0.25 \text{ rad/s}^2}$$

$$= 6.67 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

Empirical relationship : $\tau \text{ (N}\cdot\text{m)} = 6.67 \times 10^{-3} \text{ kg}\cdot\text{m}^2 \times \alpha \text{ [rad/s}^2] + 0.6 \times 10^{-3} \text{ N}\cdot\text{m}$

Measurements & calculations

Table showing the elapsed rotation time for discs of different masses to complete revolutions as well as the applied torque and angular acceleration

Mass, m [$\times 10^{-3}$ kg]	Time, t [s]	Applied torque τ [N.m] $\times 10^{-3}$	Angular acceleration α [rad/s ²]
9.5	10.25	2.33	0.359
19.5	8.06	4.78	0.580
29.5	5.96	2.7 7.23	1.061
39.5	5.79	9.68	1.125
49.5	4.87	12.13	1.5895

Conclusion

$$I = \frac{1}{2} M R^2$$
$$= 7.38 \times 10^{-3} \text{ kg.m}^2$$

There is a difference between the gradient of the graph (I) and the worked out inertia which may be due to human / experimental errors such as incorrect values as points on graph are estimations; inaccurate recording of time (ie not stopping at exactly 3 revolutions); the string also may not have been perfectly perpendicular

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(9)

The relationship between the change in applied torque (ΔT [N.m]) and the change in angular acceleration of disc ϵ

